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**CE-504 (GS)**  
**B.E. V Semester** Examination, June 2020  
**Grading System (GS)**  
**Structural Design and Drawing-I (RCC)**  
*Time : Three Hours*

**Maximum Marks : 70**

- Note :** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. Mention properties of cement concrete and reinforcing steel.
2. Discuss the merits and demerits of LSM over WSM.
3. What are various steps to be taken account in to design of one way slab? Explain.
4. What do you understand by working stress method and limit state method of design. Explain clearly.
5. Explain effective length in a column. Also explain steps in design at square footing.
6. Differential one way slab and two way slab. Also differential shallow beam and deep beam.
7. Layout a longitudinal staircase 1.25m wide for a residential flat to climb 3.2m between floors in a staircase room 4.5m×5.5m. Design the slab and detail the steel, assuming  $f_{ck}=20\text{N/mm}^2$  and  $f_y=415\text{N/mm}^2$ . Assume that the steps are built on the slab in brick work.

OR

Explain Waffle slabs and circular slab design.

8. Write short notes on any three of the following:
  - a) Balanced, under reinforced and over reinforced sections
  - b) Doubly reinforced beam
  - c) Flat slabs
  - d) Raft foundation
  - e) Slabless staircase

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